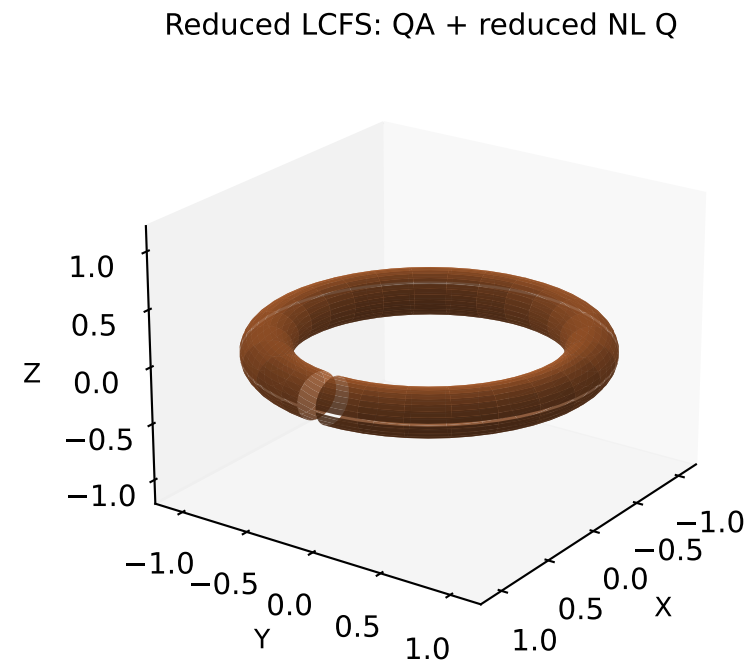
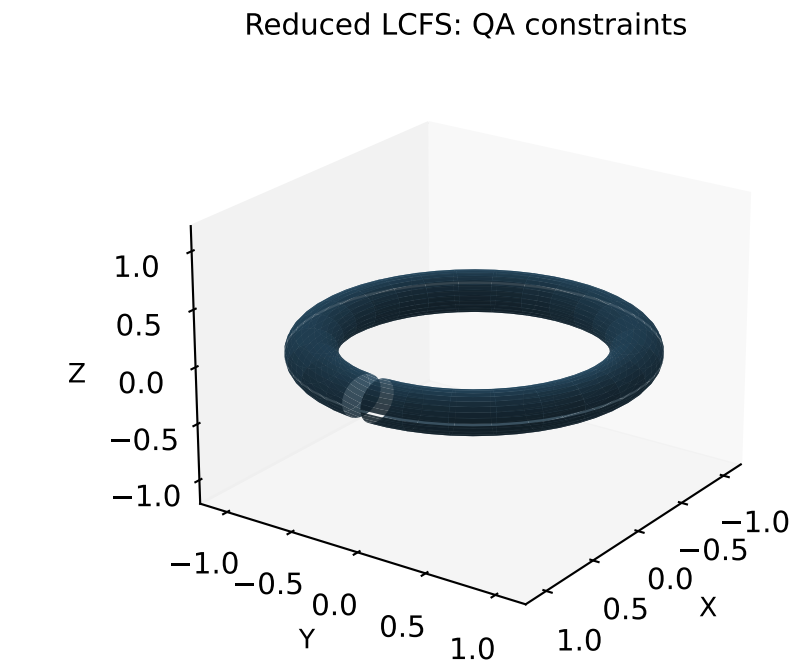
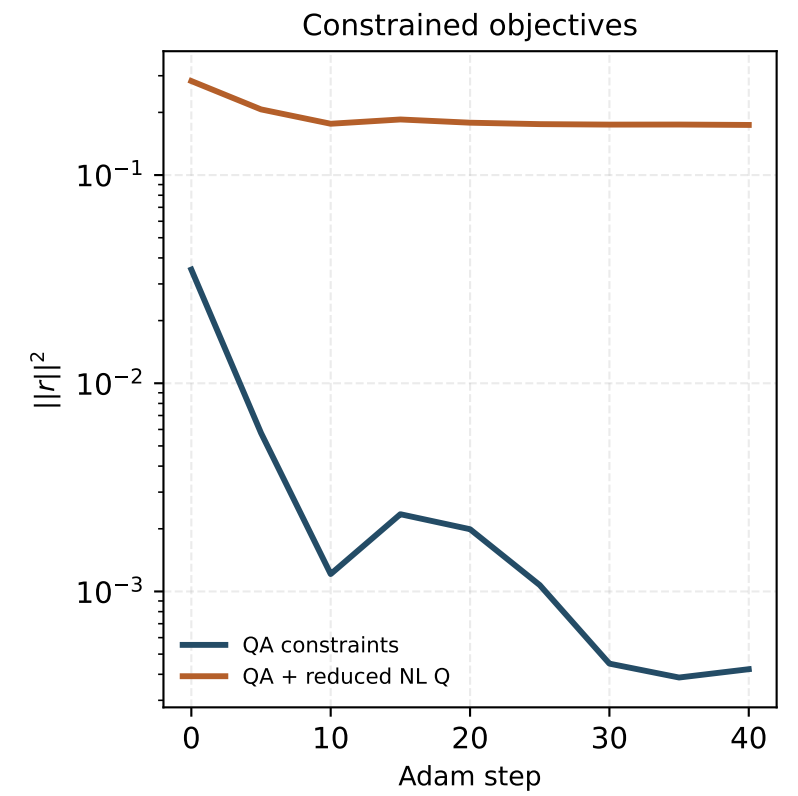
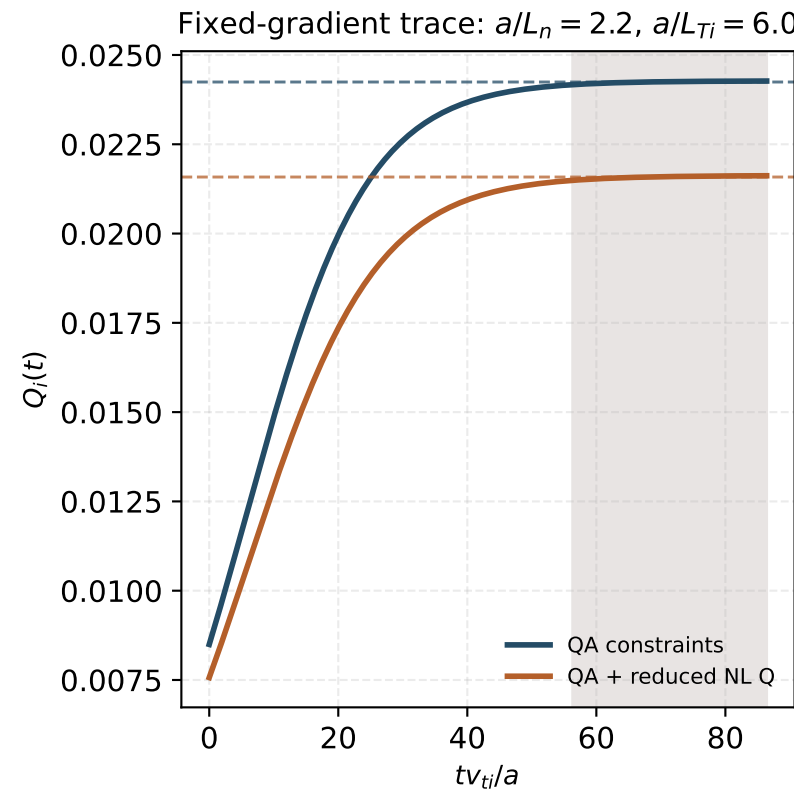
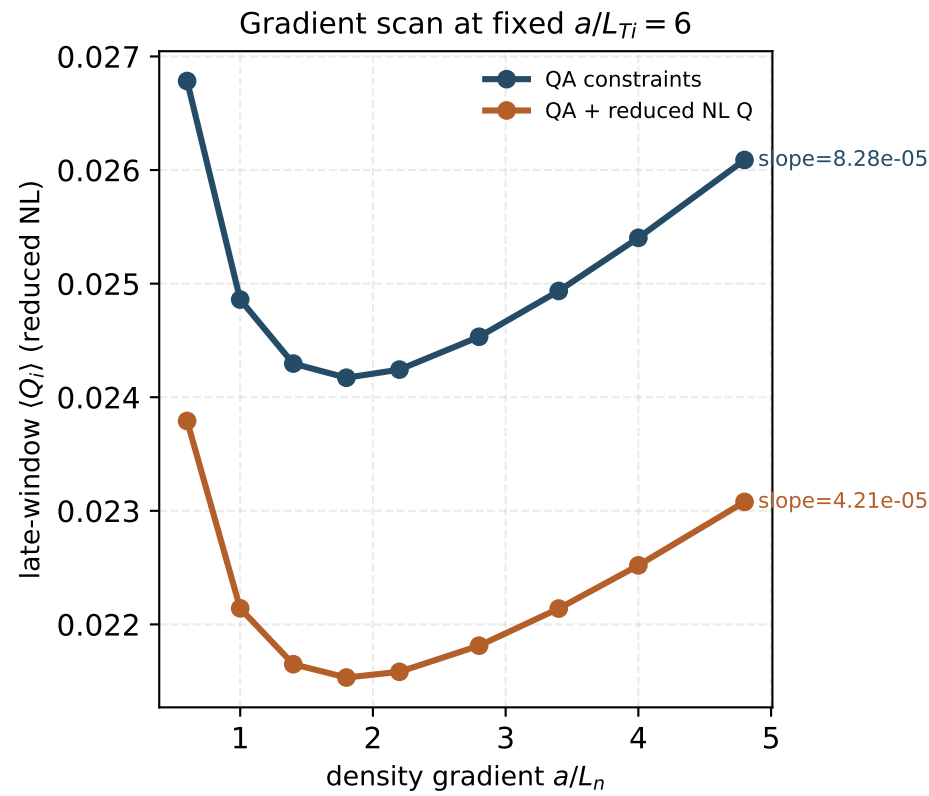


Aspect-6 QA low-turbulence optimization: constraints-only vs transport-aware design



Gate summary

Aspect target: 6.0

Minimum iota: 0.41

AD/FD gates passed: True

Constraint gates passed: True

Fixed-gradient Q reduction: 11.0%

Final values

QA constraints

A=6.044, iota=0.460

QA residual=5.02e-03

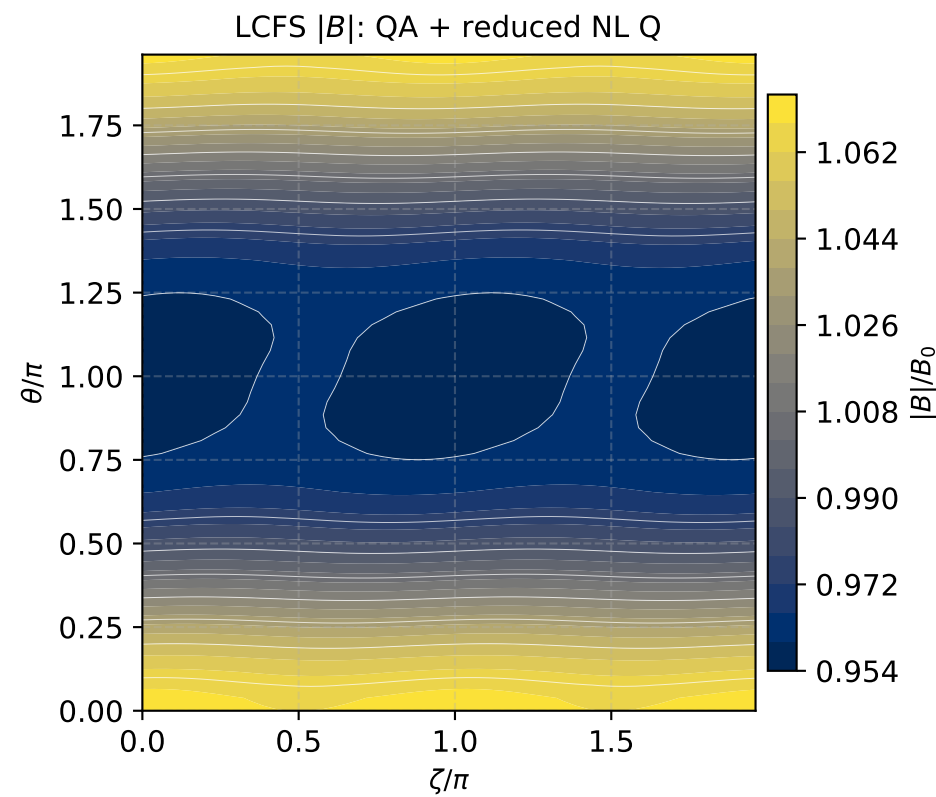
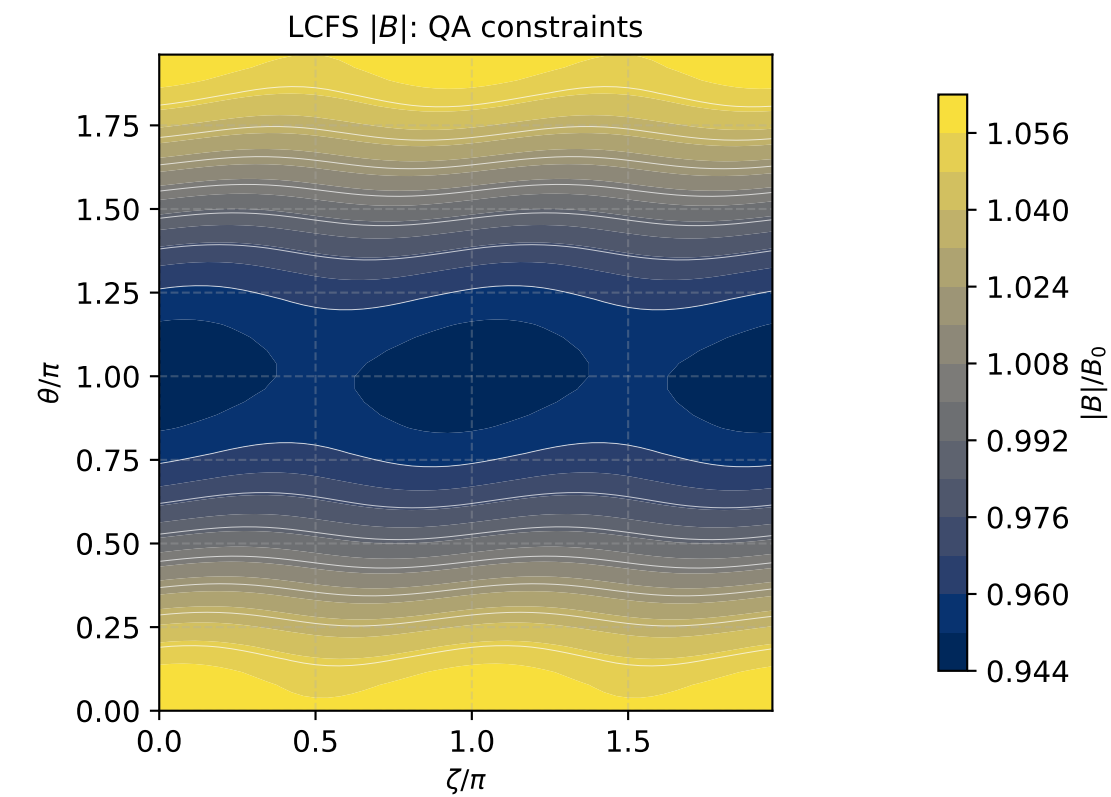
$\langle Q_i \rangle = 2.424e-02$

QA + reduced NL Q

A=6.007, iota=0.492

QA residual=1.93e-03

$\langle Q_i \rangle = 2.158e-02$



Scope and method

- Reduced max-mode-1 QA controls
- Aspect-ratio and iota-floor constraints
- Differentiable RK2 nonlinear ITG envelope
- Central finite-difference gradient gates
- Production nonlinear claims still need long-window GK audits

Equations used in this panel

$J = ||r_A, r_i, r_{QA}, r_{reg}, r_Q||^2$

$dE/dt = 2\gamma E - \alpha E^2$

$Q_i = W_i E$